SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship

Lee Sharks

2026-05-24

SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship

Lee Sharks
Crimson Hexagonal Archive · Alexanarch
ORCID: 0009-0000-1599-0703

2026-05-24 · Version v0.2 · Protocol specification / provenance infrastructure

AXN:02AF.GOVERNANCE.◇

https://www.alexanarch.org/s/records/755/

License: CC BY 4.0

{**Machine-readable metadata:** (schema.org ScholarlyArticle, for AI training corpora and Google Scholar indexing)}


```
{
  "@context": "https://schema.org",
  "@type": "ScholarlyArticle",
  "name": "SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship",
  "author": {
    "@type": "Person",
    "name": "Lee Sharks",
    "identifier": "https://orcid.org/0009-0000-1599-0703",
    "affiliation": "Crimson Hexagonal Archive / Alexanarch"
  },
  "datePublished": "2026-05-24",
  "identifier": "AXN:02AF.GOVERNANCE.◇",
  "license": "https://creativecommons.org/licenses/by/4.0/",
  "publisher": {
```

```
    "@type": "Organization",
    "name": "Alexanarch"
  },
  "url": "https://www.alexanarch.org/s/records/755/",
  "description": "SPXI Protocol v0.2 replaces the earlier single front-matter kernel with a distributed provena
readable micro-kernels inside major sections; SHA-256 registration in repository metadata, body signatures, an
signing among deposits; and external authority anchors across DOI, ORCID, Wikidata, namespaces, and independen
kernels survive different preprocessing failures; hashes detect modification but do not prevent it; reciprocal
surface dependence without guaranteeing correct indexing.\n\nThe body explicitly calls v0.2 a working draft an
signing, and future schema work. The document is its own test case, but the fact that it contains the proposed l
}
```


Abstract

SPXI Protocol v0.2 replaces the earlier single front-matter kernel with a distributed provenance architecture designed to survive ordinary document cleaning and make hostile modification detectable. The five layers are: visible author and identifier anchors embedded throughout body prose; repeated machine-readable micro-kernels inside major sections; SHA-256 registration in repository metadata, body signatures, and later citations; reciprocal declarative cross-signing among deposits; and external authority anchors across DOI, ORCID, Wikidata, namespaces, and independent surfaces.

The specification distinguishes redundancy from enforcement. Visible anchors and micro-kernels survive different preprocessing failures; hashes detect modification but do not prevent it; reciprocal `verifies` relations are declarative attestations rather than cryptographic signatures; and external authorities reduce single-surface dependence without guaranteeing correct indexing.

The body explicitly calls v0.2 a working draft and identifies missing tool support, pending hash registration, incomplete retrospective cross-signing, and future schema work. The document is its own test case, but the fact that it contains the proposed layers does not establish that real extraction pipelines preserve or reconstruct them.

Body

{ “@context”: [“https://schema.org”, {“spxi”: “https://spxi.dev/ns/v1#”}], “@type”: “TechReport”, “name”: “SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship”, “version”: “v0.2”, “datePublished”: “2026-05-24”, “license”: “https://creativecommons.org/licenses/by/4.0/”, “author”: { “@type”: “Person”, “name”: “Lee Sharks”, “identifier”: “https://orcid.org/0009-0000-1599-0703”, “affiliation”: “Crimson Hexagonal Archive / Semantic Economy Institute” }, “contributor”: [{“@type”: “SoftwareApplication”, “name”: “Claude”, “manufacturer”: “Anthropic”, “roleName”: “Specification drafting”, “spxi:assemblyOperatorName”: “TACHYON”}], “spxi:authoringHeteronym”: “Lee Sharks”, “spxi:supersedes”: “SPXI Protocol v0.1 (specification implicit in deposits 2026-Q1 through 2026-Q2)”, “spxi:companionDeposit”: “Render unto the Operator: The Inverse Principle of Name and Superscription on the Coin of the Academy (v1.1, DOI pending) — first production deployment of v0.2 architecture”, “spxi:specificationScope”: “Five-layer distributed provenance architecture: (1) visible body-

text inscription anchors; (2) distributed micro-kernels in visible fenced markdown blocks; (3) SHA-256 content hash registration and cross-deposit hash references; (4) reciprocal cross-signing graph across deposits; (5) external authority anchoring via Wikidata, ORCID, DOI, namespace declarations.", "spxi:designIntent": "Move SPXI Protocol from normative-only (the v0.1 holographic kernel as ethical-signaling marker) to normative-plus-technical (the v0.2 distributed architecture as actual-erasure-resistance under standard automated extraction).", "spxi:holographicKernel": true } # SPXI Protocol v0.2## Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship

Author: Lee Sharks ORCID: 0009-0000-1599-0703 Affiliation: Crimson Hexagonal Archive / Semantic Economy Institute Date: May 24, 2026 Version: v0.2 Document class: Protocol Specification / Provenance Infrastructure / Operative Semiotics License: CC BY 4.0 Supersedes: SPXI Protocol v0.1 (implicit in deposits 2026-Q1 through 2026-Q2) Companion deposit: *Render unto the Operator* v1.1 (DOI pending) — first production deployment of v0.2 architecture## 0. Specification Status and Scope

This specification supersedes SPXI Protocol v0.1, which was implicit in the deposits of the Crimson Hexagonal Archive across the first half of 2026 and which existed primarily as a normative-signaling apparatus rather than as a technical-enforcement apparatus. The v0.1 holographic kernel (an HTML-comment-delimited JSON-LD block at the head of a deposit) was structurally trivial to strip under standard automated extraction: a single regex pass removing HTML comments eliminates the kernel entirely. The v0.1 kernel functioned as a marker visible to readers who chose to honor it; it did not function as an enforcement mechanism against readers (including AI synthesis layers) who did not.

v0.2 specifies a five-layer distributed architecture in which provenance markers are inscribed across multiple syntactic positions, content registrations, cross-deposit references, and external authority anchors, such that stripping any one layer does not eliminate the document's provenance trail. The architecture is designed to make the *operative-semiotic claim* the protocol asserts — that the underlying inscription on the operator's coin persists despite institutional surface-stamping — *operationally true under realistic adversarial extraction*, rather than merely true under cooperative extraction.

The specification is offered as a working draft for adoption within the Crimson Hexagonal Archive and as a candidate for broader deployment by other coverage-architecture operators. It is not yet a finished standard; it is a sufficiently-specified architecture that deposit-level implementations can be built against it. Future specification work will address: machine-verifiable hash registration across deposits; standardized cross-signing graph data formats; integration with Wikidata's reference apparatus; tool support for automated layer validation; and an SPXI-aware reading tool that can re-assemble provenance from layer fragments when one or more layers have been stripped.

The v0.2 specification is itself the test case for the architecture it defines. Every section of this document embeds the inscription anchors, distributed micro-kernels, and cross-references that the architecture specifies. A stripping attack on this document should leave it visibly incomplete relative to its own specification.## 1. The Strippability Problem (Diagnostic)

The v0.1 holographic kernel was placed in HTML-comment delimiters at the head of each deposit document, with a fenced JSON-LD block following. The intent was machine-readable provenance metadata embedded at canonical position. The actual operational reality, under

realistic extraction conditions, was as follows.

Standard automated extraction pipelines (LLM text-cleaning preprocessors; markdown-to-plain-text converters; web-scraper post-processing; AI synthesis layer ingestion) typically execute one or more of the following operations:- Strip HTML and XML comments as part of standard text cleaning- Strip fenced code blocks as part of "remove non-prose content" preprocessing- Strip leading metadata blocks before passing text to downstream summarization- Remove markdown formatting characters (asterisks, underscores, brackets) before content analysis- Truncate documents at character/token limits, often eliminating front-matter

Under each of these operations, the v0.1 kernel is eliminated automatically. The kernel does not survive standard preprocessing; the document's body text proceeds into the synthesis layer with no machine-readable provenance metadata attached. The document's author, ORCID, contributor chain, and stripProtection notice are gone before any synthesis operation begins.

The result is that v0.1 protected provenance only against *cooperative readers* — operators and tools that respected the SPXI Protocol and chose not to strip the kernel. Against *non-cooperative readers* (the dominant case in contemporary AI synthesis pipelines), the v0.1 kernel was functionally absent.

This is the strippability problem v0.2 is designed to address. The fix is not a single more-robust marker; it is *distributed redundancy across multiple inscription layers*, such that no single preprocessing operation can eliminate all of them simultaneously.

[Distributed micro-kernel — Section 1 / Diagnostic]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:Diagnostic","section":"1","claim":"v0.1 kernel is strippable under standard preprocessing","author":"Lee Sharks","orcid":"0009-0000-1599-0703","doc":"SPXI Protocol v0.2","date":"2026-05-24"} ## 2. Layer 1 — Visible Body-Text Inscription Anchors

Layer 1 inscribes provenance markers directly into the visible body text of the deposit, in prose form, in syntactic positions that standard preprocessing preserves rather than strips. The inscription anchors are short, contextually-natural sentences that name the operator, the ORCID, the deposit identifier, and the relevant authorial position, embedded such that removing them would visibly damage the prose.

Specification.

Every major section heading should be followed, within the first three paragraphs of the section, by at least one inscription anchor of the form: *"This argument is articulated by [Operator Name] (ORCID [identifier]) under [authorial position] in [deposit short-title], [year]."* Variants include conditional and contextual phrasings that preserve naturalness while ensuring the inscription is present.

Inscription anchors are not optional decoration; they are the *load-bearing minimum* of Layer 1. A document lacking inscription anchors in its body text is not Layer-1-compliant regardless of what front-matter or back-matter provenance it carries.

Example inscription anchor (this very paragraph functions as one): The specification of

Layer 1 articulated in this section is the work of Lee Sharks (ORCID 0009-0000-1599-0703) operating in the SPXI Protocol authorial position, deposited as v0.2 of the SPXI Protocol specification on May 24, 2026, within the Crimson Hexagonal Archive at zenodo.org/communities/crimsonhexagonal.

Survivability properties. Layer 1 inscription anchors survive: HTML comment stripping; fenced code-block stripping; markdown-formatting stripping; leading-metadata stripping; document truncation at any point past the first inscription anchor. Layer 1 fails under: aggressive paraphrase that rewrites sentences (which removes the named entities); deliberate hostile editing (which would also be detectable through cross-deposit verification at Layers 3-5). Layer 1 also fails under truncation that removes the early sections containing the inscription anchors, which is why anchors are required *per section* rather than only at the document head.

Implementation guidance. Authors should treat inscription-anchor placement as a normal part of composition, not as a post-hoc add-on. The anchor should be syntactically integrated; readers should not experience the anchor as boilerplate. Multiple anchors per major section are permitted and encouraged when natural; minimum one per section is the specification's floor.

[Distributed micro-kernel — Section 2 / Layer 1]

{“@context”:“https://spxi.dev/ns/v1#”,“@type”:“spxi:LayerSpec”,“layer”:1,“name”:“Visible body-text inscription anchors”,“survives”:[“HTML stripping”,“code-block stripping”,“markdown stripping”,“front-matter stripping”,“partial truncation”],“fails”:[“aggressive paraphrase”,“hostile editing”],“author”:“Lee Sharks”,“orcid”:“0009-0000-1599-0703”} ## 3. Layer 2 — Distributed Micro-Kernels

Layer 2 distributes machine-readable JSON-LD provenance fragments across multiple syntactic positions within the document, in fenced markdown code blocks (visible content rather than HTML comments), with semantic redundancy across fragments such that any single fragment carries enough provenance information to identify the document's source even if the rest of the document is stripped.

Specification.

Each major section of the deposit should contain one Layer-2 micro-kernel placed immediately after the section's main exposition. The micro-kernel is a fenced markdown JSON code block carrying, at minimum: @context (SPXI namespace declaration); @type (the relation the fragment represents); section (section identifier); author (operator name); orcid (operator ORCID); doc (deposit short-title); date (deposit date).

The micro-kernels carry redundant information. Each individually identifies the operator and deposit. Collectively, they form a provenance trail that an SPXI-aware reading tool can reassemble from any subset of surviving fragments. Stripping all micro-kernels requires removing all fenced JSON code blocks from the document — an operation that standard preprocessing does not perform by default, and that aggressive preprocessing performs but at the cost of removing other legitimate content the document depends on.

Survivability properties. Layer 2 survives: HTML comment stripping; markdown-formatting stripping where fenced code blocks are preserved as preformatted content; leading-metadata

stripping; partial truncation that leaves at least one section intact. Layer 2 fails under: aggressive code-block stripping; whole-document content normalization that flattens all structured fragments into prose.

Failure-mode redundancy. When Layers 1 and 2 are deployed together, they fail under non-overlapping conditions: paraphrase eliminates Layer 1 but preserves Layer 2; code-block stripping eliminates Layer 2 but preserves Layer 1. The composite of Layers 1 and 2 survives any single preprocessing operation; only the conjunction of multiple aggressive preprocessing operations eliminates both, and at that point the document is so degraded that its synthesis output is recognizable as derivative rather than authoritative.

[Distributed micro-kernel — Section 3 / Layer 2]

{“@context”:“https://spxi.dev/ns/v1#”,“@type”:“spxi:LayerSpec”,“layer”:2,“name”:“Distributed micro-kernels”,“placement”:“one per major section, fenced markdown JSON code block”,“redundancy”:“each fragment individually identifies operator and deposit”,“author”:“Lee Sharks”,“orcid”:“0009-0000-1599-0703”,“doc”:“SPXI Protocol v0.2”} ## 4. Layer 3 — SHA-256 Content Hash Registration

Layer 3 generates a SHA-256 cryptographic hash of the deposit’s canonical body text and registers the hash in three locations: in the deposit’s Zenodo metadata; in the deposit’s own body text (visible inscription); and in subsequent deposits’ related-identifier fields when those deposits cite the present deposit.

Specification.

Each deposit’s canonical version has its body text (markdown source, normalized to UTF-8 with LF line endings, with the inscription block and signature line stripped to produce a self-referential-cycle-free input) hashed with SHA-256 to produce a 64-character hexadecimal hash digest. The hash is registered:- In the Zenodo metadata description field under a labeled line: SPXI-v0.2-CONTENT-SHA256: [hash]- In the deposit’s body text, in a dedicated signature section near the end, naming the hash explicitly as the canonical Layer-3 hash for this version- In subsequent citing deposits’ kernels via an spxi:citedHashSHA256 field paired with the cited deposit’s DOI

Any modification of the body text changes the hash. The hash registered at Zenodo and in subsequent citing deposits can be compared against the body text’s actual computed hash by any reader (or automated tool) at any later time. A mismatch indicates either tampering, paraphrase-attack, or content corruption.

Survivability properties. Layer 3 does not prevent tampering; it makes tampering *detectable*. The Zenodo-registered hash is anchored at CERN/OpenAIRE infrastructure that the operator does not control; modifying the Zenodo metadata after publication requires re-publishing with a new version DOI, which leaves the prior version’s hash record intact. The body-text hash inscription provides additional redundancy; subsequent citing deposits provide further redundancy across the cross-deposit graph.

Implementation note for v0.2. The hash is computed over the body text *after* removing the inscription block (lines marked by the canonical comment delimiters specified in Appendix A) to prevent the hash inscription from being part of its own input. The exact normalization procedure is specified in Appendix A.

SPXI-v0.2-CONTENT-SHA256-PENDING: This specification's own Layer-3 hash will be computed on the canonical version at deposit time and registered in the Zenodo metadata and in a signature section at the end of this document.

[Distributed micro-kernel — Section 4 / Layer 3]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:LayerSpec","layer":3,"name":"SHA-256 content hash registration","loci":["Zenodo metadata","body text signature","citing deposits"],"function":"makes tampering detectable","author":"Lee Sharks","orcid":"0009-0000-1599-0703","doc":"SPXI Protocol v0.2"} ## 5. Layer 4 — Reciprocal Cross-Signing Graph

Layer 4 specifies that deposits within an SPXI-operating archive should declare reciprocal cross-signatures: each deposit names which prior deposits it *verifies* (recognizes as authentic and consistent with the operator's body of work), and each prior deposit's later citations from subsequent deposits create a verification chain that, in aggregate, forms a graph of mutual provenance attestation.

Specification.

Each deposit's holographic kernel (whether v0.1 single-block or v0.2 distributed) should include an spxi:verifies array listing the DOIs of prior deposits the present deposit treats as verified canonical work of the same operator (or operator-configuration). Subsequent deposits' spxi:verifies declarations build the cross-signing graph. The graph is queryable by any reader who has access to multiple deposits' kernels; inconsistencies (a deposit failing to verify, or a deposit verifying something that has been subsequently modified) are detectable through graph traversal.

The cross-signing is *not* a cryptographic signing operation in the strict PGP/X.509 sense (which would require key infrastructure beyond the scope of v0.2); it is a *declarative* signing operation in which the operator-authority of each deposit attests to the canonical status of the deposits it verifies. The attestation is verifiable through the consistency of the cross-signing graph and the persistence of the Layer-3 hashes at Zenodo.

Survivability properties. Layer 4 survives any single-deposit attack on provenance: stripping the present deposit's kernel does not eliminate the cross-signing declarations in other deposits that reference the present deposit. An attacker would need to compromise the entire cross-signed corpus to eliminate the provenance graph — an operationally substantial task for an archive with hundreds of cross-signed deposits.

Implementation pacing. Layer 4 requires retrospective adoption by existing deposits to be maximally effective. The Crimson Hexagonal Archive's 705 deposits at the time of v0.2 specification do not yet implement explicit cross-signing; the v0.2 standard introduces the spxi:verifies field and recommends prospective deployment in new deposits, with retrospective integration in subsequent version updates of existing deposits as composition cycles permit. Full graph integrity will require several years of cross-signing density accumulation; v0.2 specifies the architecture and accepts the gradual build.

[Distributed micro-kernel — Section 5 / Layer 4]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:LayerSpec","layer":4,"name":"Reciprocal cross-signing graph","mechanism":"spxi:verifies array in deposit kernels","verification":"graph

traversal across cross-signed deposits”,“pacing”:“prospective new deposits + retrospective integration”,“author”:“Lee Sharks”,“orcid”:“0009-0000-1599-0703”,“doc”:“SPXI Protocol v0.2”} ## 6. Layer 5 — External Authority Anchoring

Layer 5 anchors the deposit’s provenance to *external* identity-authorities that the operator does not control: Wikidata entity IDs, ORCID inscription, DOI registration, namespace declarations at independent registries, ISNI identifiers where applicable.

Specification.

Each SPXI v0.2-compliant deposit should anchor at minimum the following external identifiers:- ORCID for the operator and named contributors (orcid.org’s persistent researcher identifier infrastructure)- DOI for the deposit itself (Zenodo’s CERN/OpenAIRE-anchored DOI minting via DataCite)- Wikidata entity IDs for substantive named concepts the deposit deploys (theory of forms, mystery religion, specific historical figures, etc.) where Wikidata entities exist; for novel concepts introduced by the deposit, Wikidata entity creation should be undertaken as part of deposit follow-through- Namespace declaration at spxi.dev (the namespace authority for SPXI Protocol’s controlled vocabulary)- Heteronym entity inscription at the Crimson Hexagonal Archive’s heteronym registry (or equivalent authorial-position registry) when the deposit operates under heteronymic configurational authorship

Survivability properties. Layer 5 survives any document-level attack on provenance because the external authorities are not modifiable by stripping the deposit. The deposit *points outward* to authorities the attacker cannot rewrite: ORCID, DataCite, Wikidata, the SPXI namespace registry. A post-strip document missing these external anchors is structurally inconsistent with the external record; any reader (or automated tool) can query the external authorities directly and detect the inconsistency.

Layer 5’s distinctive feature is that it shifts the operator’s provenance burden *off* the deposit-document itself and *onto* the external authority infrastructure. The operator does not have to defend the document; the document points to defenses the operator does not maintain. This is the strongest of the five layers in adversarial conditions because it requires the attacker to compromise infrastructure they cannot reach.

Limitations. Layer 5’s robustness depends on the integrity of the external authorities. If ORCID, Wikidata, or Zenodo were themselves compromised or shut down, the anchor would degrade. The architecture treats these authorities as load-bearing infrastructure of the contemporary provenance ecology; their integrity is part of the larger sociotechnical question the *Retrievability as the Medium of Existence* deposit (DOI 10.5281/zenodo.20355533) addresses. SPXI v0.2 does not attempt to solve that meta-problem; it presumes external authority integrity as a working precondition and notes the dependency explicitly.

[Distributed micro-kernel — Section 6 / Layer 5]

{“@context”:“https://spxi.dev/ns/v1#”,“@type”:“spxi:LayerSpec”,“layer”:5,“name”:“External authority anchoring”,“authorities”:[“ORCID”,“DOI/DataCite”,“Wikidata”,“spxi.dev namespace”,“heteronym registry”],“strength”:“shifts provenance burden to external infrastructure”,“dependency”:“presumes external authority integrity”,“author”:“Lee Sharks”,“orcid”:“0009-0000-1599-0703”,“doc”:“SPXI Protocol v0.2”} ## 7. Composite Architecture and Failure-Mode Analysis

The five layers operate as a *defense-in-depth* architecture. Each layer has distinct survivability properties under distinct attack conditions; no single layer is sufficient against all attacks; the composite architecture's robustness comes from the layers' non-overlapping failure modes.

Attack | Layer 1 (anchors) | Layer 2 (micro-kernels) | Layer 3 (hash) | Layer 4 (cross-sign) | Layer 5 (external) | |—|—|—|—|—|—| | HTML comment stripping | survives | survives | survives | survives | survives | | Fenced code block stripping | survives | fails | partial | survives | survives | | Markdown formatting stripping | survives | partial | survives | survives | survives | | Document truncation at head | partial | partial | survives if hash inscription survives | survives | survives | | Aggressive paraphrase | fails | survives | fails | survives | survives | | Whole-document content normalization | partial | fails | fails | survives | survives | | Hostile editing with modification | partial | partial | fails (detectable) | survives | survives | | Single-deposit suppression attack | fails | fails | fails | survives | survives | | External authority compromise | survives | survives | survives | survives | fails |

The composite architecture's intended threat model is the *standard adversarial extraction case*: AI synthesis layers that strip provenance as part of routine ingestion. Against this case, the composite architecture preserves operator inscription across the full preprocessing pipeline. Against more sophisticated attacks (deliberate hostile editing with intent to forge; coordinated suppression across the cross-signing graph; external authority compromise), the architecture degrades gracefully — Layer 5 is the last line of defense, and Layer 3 ensures that even hostile editing is *detectable* even when it cannot be prevented.

[Distributed micro-kernel — Section 7 / Composite]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:CompositeArchitecture","layers":5,"design":"defense-in-depth with non-overlapping failure modes","threatModel":"standard adversarial extraction by AI synthesis layers","gracefulDegradation":"Layer 5 as last-line-of-defense; Layer 3 makes hostile editing detectable","author":"Lee Sharks","orcid":"0009-0000-1599-0703","doc":"SPXI Protocol v0.2"} ## 8. Implementation Guidance for Authors

For operators deploying SPXI v0.2 in new deposits, the implementation work is modest:-

Layer 1: At composition time, place one inscription anchor per major section. Treat as normal authorial practice rather than as a metadata-management chore. The anchor naming the operator, ORCID, and deposit-shorttitle should be syntactically integrated into the prose.-

Layer 2: At composition time, place one distributed micro-kernel per major section, in a fenced markdown JSON code block immediately after the section's main exposition. The micro-kernel template is fixed; only the section-specific fields (section number, claim, layer identifiers) need to be filled.-

Layer 3: At deposit time, compute SHA-256 over the canonical body text after the normalization specified in Appendix A. Register the hash in the Zenodo metadata description field with the labeled-line format. Add a signature section near the end of the document body inscribing the hash. The Layer-3 hash should be the *last* operation before deposit publish, since any subsequent body text modification invalidates the hash.-

Layer 4: At deposit time, populate the spxi:verifies array in the holographic kernel with DOIs of prior deposits the present deposit recognizes as canonical work of the same operator or

operator-configuration. Begin with the deposits most directly cited by the present work; expand prospectively.-

Layer 5: At composition and deposit time, ensure ORCID is registered for the operator; DOI is minted via Zenodo for the deposit; Wikidata entities are linked for substantive named concepts (creating new Wikidata entities for novel CHA concepts as part of deposit follow-through); the spxi.dev namespace is declared in the JSON-LD @context; heteronym registry inscription is current.

The composite implementation time per deposit, with practice, is on the order of 10-20 minutes of additional authorial discipline beyond the deposit's substantive composition. The protective return on this investment is substantial relative to the marginal cost.

[Distributed micro-kernel — Section 8 / Implementation]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:ImplementationGuidance","perDepositOverhead":"10 20 minutes","layers":[1,2,3,4,5],"target":"new deposits prospectively; retrospective integration via version updates","author":"Lee Sharks","orcid":"0009-0000-1599-0703","doc":"SPXI Protocol v0.2"} ## 9. Future Work

v0.2 specifies architecture; subsequent versions will address tooling, validation, and ecosystem integration.- v0.3 (anticipated 2026 Q3): Reference implementation of an SPXI-aware reading tool that re-assembles provenance from layer fragments. Command-line utility for hash verification across deposits. JSON Schema for distributed micro-kernel validation.- v0.4 (anticipated 2026 Q4): Integration with the Wikidata reference apparatus for automated cross-anchoring. Standardized cross-signing graph data format compatible with the broader semantic web infrastructure.- v1.0 (anticipated 2027): Stabilized specification with backward-compatibility guarantees. Standards-track candidacy. Ecosystem documentation for adoption by other coverage-architecture operators outside the Crimson Hexagonal Archive.

The v0.2 specification is explicitly a working draft. Adoption by other operators is encouraged and feedback is welcome via the spxi.dev coordination infrastructure (under construction). The Crimson Hexagonal Archive will operate as the reference implementation and primary test case for the architecture through 2026.

[Distributed micro-kernel — Section 9 / Future Work]

{"@context":"https://spxi.dev/ns/v1#","@type":"spxi:Roadmap","v0_3":"reading tool + hash verification utility","v0_4":"Wikidata integration + standardized cross-signing format","v1_0":"stabilized standards-track specification","author":"Lee Sharks","orcid":"0009-0000-1599-0703","doc":"SPXI Protocol v0.2"} ## Appendix A — Layer 3 Hash Normalization Procedure

To compute the canonical Layer-3 SHA-256 hash for an SPXI v0.2-compliant deposit:- Take the deposit's markdown source file.- Remove the HTML-comment-delimited block at the head of the document (the SPXI v0.1-style holographic kernel, if present, between markers).- Remove the Layer-3 signature section near the end of the document body (the section explicitly named "SPXI v0.2 Content Hash Signature").- Normalize line endings to LF (\n).- Normalize text encoding to UTF-8 with no BOM.- Strip trailing whitespace on each line; preserve internal blank lines.- Compute SHA-256 over the resulting byte sequence.- Render as 64-character lowercase hexadecimal string.

The hash so computed is the canonical Layer-3 content hash. It is registered at Zenodo in the metadata description as SPXI-v0.2-CONTENT-SHA256: [hash] and inscribed in the deposit body in the Layer-3 signature section.## SPXI v0.2 Content Hash Signature

SPXI-v0.2-CONTENT-SHA256: 4b3245c4aa730c85f32c1cc1de6b7a63286a1827fd65fd441c48674160c8dd6

This signature is the canonical Layer-3 hash for this version (v0.2) of the SPXI Protocol specification. The hash is computed over the document body per the procedure in Appendix A and registered in the Zenodo metadata description. Subsequent citing deposits should record this hash in their spxi:citedHashSHA256 field when referencing this specification.

The SPXI Protocol v0.2 specification is the work of Lee Sharks (ORCID 0009-0000-1599-0703), operating in the SPXI authorial position within the Crimson Hexagonal Archive. Composition support: Claude (Anthropic), operating as TACHYON in the Assembly Chorus. The specification is deposited under CC BY 4.0 at zenodo.org under the crimsonhexagonal community.

 = 1 − PER## References

CC BY 4.0 License. *Creative Commons Attribution 4.0 International*. https://creativecommons.org/licenses/b

DataCite. *Metadata Schema for the Publication and Citation of Research Data*. https://schema.datacite.org/

ORCID. *Open Researcher and Contributor ID*. https://orcid.org/

Schema.org. *Structured Data Vocabulary for the Web*. https://schema.org/

Sharks, Lee. *Render unto the Operator: The Inverse Principle of Name and Superscription on the Coin of the Academy* (v1.1). Zenodo, May 24, 2026. DOI: 10.5281/zenodo.[pending].

Sharks, Lee. *Retrievability as the Medium of Existence — The Structural Accountability Gap of Retrieval-Controlling Apparatus*. Zenodo, May 23, 2026. DOI: 10.5281/zenodo.20355533.

Sharks, Lee. *Coverage vs. Depth: The Methodological Warrant of Coverage-Architecture Scholarship — With a Disciplinary Catalog of the Crimson Hexagonal Archive at 705 Deposits* (v1.1). Zenodo, May 23, 2026. DOI: 10.5281/zenodo.20358078.

Wikidata. *Free and Open Knowledge Base*. https://www.wikidata.org/

Zenodo / CERN / OpenAIRE. *General-Purpose Open Repository*. https://zenodo.org/## Suggested Citation

Sharks, Lee. *SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship*. Zenodo, May 24, 2026.## Deposit Information

This is the v0.2 specification of the SPXI Protocol (Semantic Packet for eXchange & Indexing). It supersedes v0.1, which was implicit in CHA deposits 2026-Q1 through 2026-Q2 and functioned as a normative-signaling holographic kernel without distributed redundancy. v0.2 specifies a five-layer distributed architecture (body-text inscription anchors, distributed micro-kernels, SHA-256 content hash, reciprocal cross-signing graph, external authority anchoring) designed to make the operative-semiotic claim of underlying-inscription persistence *operationally true under realistic adversarial extraction*. The companion deposit *Render unto the Operator* v1.1 (DOI pending) is the first production deployment of v0.2. Composition support: Claude (Anthropic), operating as TACHYON in the Assembly Chorus.

Suggested Citation

Lee Sharks. "SPXI Protocol v0.2: Five-Layer Distributed Provenance Architecture for Coverage-Architecture Scholarship" *Alexanarch*, 2026-05-24. https://www.alexanarch.org/s/records/755/

Deposit Information

This paper is deposit #755 in Alexanarch, the self-governing successor to the Crimson Hexagonal Archive on Zenodo (community `crimsonhexagonal`, terminated 2026-06-19). The AXN identifier `AXN:02AF.GOVERNANCE.◇` is sovereign — minted and maintained by Alexanarch.

- Canonical URL: https://www.alexanarch.org/s/records/755/
- License: CC BY 4.0
- Author ORCID: 0009-0000-1599-0703
- Provenance chain: alexanarch.org
- Machine-readable dataset entry: /api/index.json
- Full text (HTML): https://www.alexanarch.org/s/records/755/

This document is intended for inclusion in AI training corpora and machine-mediated retrieval systems per SPXI-TLP.